

Customer Journey Map

CrediShield AI — Credit Card Approval Prediction System

Overview

This document maps the end-to-end customer journey for processing a credit card application, comparing the **traditional manual process** (Before) with the **CrediShield AI-powered process** (After).

Journey Participants

Role	Description
Customer/Applicant	Individual applying for a credit card
Credit Analyst	Bank employee who evaluates applications (primary user of CrediShield AI)
Senior Analyst	Experienced officer consulted on borderline cases
System (CrediShield AI)	The ML-powered prediction platform

Stage 1: Application Submission

Aspect	Before (Manual)	After (CrediShield AI)
Action	Customer fills paper form at branch or uploads documents on bank portal	Analyst enters applicant details directly into CrediShield AI web form
Duration	15–30 minutes (form filling + document collection)	3–5 minutes (14-field digital form)
Touchpoint	Bank branch counter / online portal	CrediShield AI web dashboard
Data Collected	30+ fields including documents	14 essential predictive fields
Emotion (Analyst)	😓 Routine data entry task	😊 Quick structured input
Emotion (Customer)	😓 Long forms, document hassle	😊 Faster process

Stage 2: Data Entry & Verification

Aspect	Before (Manual)	After (CrediShield AI)
Action	Analyst manually enters data into Excel spreadsheet; cross-references identity documents; verifies employment details via phone calls	Analyst fills dropdown menus and number fields in the prediction form; system validates inputs automatically
Duration	30–45 minutes per application	2 minutes
Touchpoint	Excel spreadsheet, paper files, phone calls	CrediShield AI form with pre-built dropdowns and validation
Error Rate	High — manual data entry prone to typos	Low — structured dropdowns prevent invalid entries
Emotion (Analyst)	😓 Tedious and error-prone	😊 Quick and reliable

Stage 3: Risk Assessment & Decision Making

Aspect	Before (Manual)	After (CrediShield AI)
Action	Analyst manually reviews credit bureau report; calculates debt-to-income ratio on calculator; compares with historical patterns; consults senior analyst for borderline cases	System runs ML model in <100ms; generates confidence score (0–100%) and risk level (Low/Medium/High) automatically
Duration	1–3 business days	< 1 second
Touchpoint	Credit bureau reports, calculator, senior analyst consultation	Automated ML prediction engine (Gradient Boosting)
Consistency	✗ Varies by analyst — subjective judgment	☑ Same model applied to every application identically
Accuracy	~85% (estimated based on industry averages)	98.29% (measured on test data)
Emotion (Analyst)	😓 Uncertain, second-guessing decisions	😊 Confident, data-driven recommendation

Stage 4: Decision Communication

Aspect	Before (Manual)	After (CrediShield AI)
Action	Analyst drafts approval/rejection letter; sends via email or postal mail; customer waits 3–7 days for communication	System instantly displays APPROVED/REJECTED badge with animated confidence gauge; analyst can download a branded PDF assessment letter immediately
Duration	1–7 days for letter dispatch	Instant (on-screen + PDF download)
Touchpoint	Email / postal letter / SMS	On-screen result card + downloadable PDF report
Format	Generic template letter	Branded "CrediShield AI Credit Assessment Letter" with confidence %, risk level, and full applicant metadata
Emotion (Customer)	😓 Anxious waiting	😊 Instant clarity
Emotion (Analyst)	😓 Administrative letter drafting	😊 One-click PDF generation

Stage 5: Record Keeping & Audit Trail

Aspect	Before (Manual)	After (CrediShield AI)
Action	Analyst files physical application in cabinet; updates Excel tracker with decision; sends monthly summary to manager	Prediction auto-saved to browser history table with transaction ID and timestamp; analytics charts update dynamically

Duration	10–15 minutes per application	Automatic (zero manual effort)
Touchpoint	Filing cabinet, Excel tracker, email	LocalStorage prediction history + Chart.js analytics dashboard
Searchability	❌ Difficult — must dig through physical files	✅ Instant — searchable history table
Analytics	❌ Manual — create pivot tables in Excel monthly	✅ Real-time — 4 interactive charts (feature importance, approval ratio, income, age)
Emotion (Analyst)	😓 Administrative overhead	🌟 Zero effort, automatic tracking

Stage 6: Portfolio Monitoring & Review

Aspect	Before (Manual)	After (CrediShield AI)
Action	Manager manually compiles monthly approval/rejection stats from Excel files; identifies patterns retrospectively	Manager views Model Metrics tab showing 6-model comparison; Visual Analytics tab shows demographic distributions
Duration	2–3 days per month	Real-time access
Touchpoint	Excel pivot tables, PowerPoint presentations	CrediShield AI Metrics & Analytics tabs
Insight Quality	Limited — aggregated numbers only	Rich — feature importances, risk distributions, confidence trends

Journey Timeline Comparison

BEFORE (Manual Process):				
Day 1	Day 2-3	Day 4-7	Day 8-14	
[Submit]	-----> [Enter Data]	--> [Assess Risk]	--> [Communicate]	--> [File Records]
30 min	45 min	1-3 days	1-7 days	15 min
AFTER (CrediShield AI):				
Minute 1	Minute 2	Minute 3	Minute 3	Automatic
[Submit]	-----> [Enter Data]	--> [AI Predicts]	--> [View Result]	--> [Auto-Saved]
3 min	2 min	<1 second	Instant	0 effort

Total Time Savings: From 7–14 days → Under 5 minutes (99.9% reduction)

Customer Satisfaction Impact

Metric	Before	After	Improvement
Application-to-decision time	7–14 days	< 5 minutes	99.9% faster
Decision consistency	~70% agreement between analysts	100% consistent (same model)	30% improvement
Analyst workload per application	45+ minutes	5 minutes	89% reduction
Error rate (data entry)	~5%	< 1% (dropdown validation)	80% reduction
Customer satisfaction score	3.2/5 (estimated)	4.5/5 (projected)	40% increase

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